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Math 208 Final Exam Term 222

Original exam date: 18 May 2023. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. The solution of $X' = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} X,\ X(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ at $t = \pi$ is equal to

(a) $\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{4\pi}$

(b) $\begin{pmatrix} -2 \\ 2 \end{pmatrix} e^{4\pi}$

(c) $\begin{pmatrix} -1 \\ -2 \end{pmatrix} e^{4\pi}$

(d) $\begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{4\pi}$

(e) $\begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{4\pi}$

2. The differential equation $x^{(4)} + 6x'' - 3x' + x = \cos 3t$ can be transferred into a system of linear first-order differential equations of the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}' = A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \cos 3t \end{bmatrix}$$

where $A =$

(a) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 3 & 6 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 3 & 6 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & -3 & -6 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -3 & -6 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 3 & -6 & 0 \end{bmatrix}$

3. Let $F(t) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $e^{At} = \begin{bmatrix} 1+2t & -4t \\ t & 1-2t \end{bmatrix}$. A particular solution for the system $X' = AX + F(t)$ with $X_p(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ will satisfy $X_p(1) =$

(a) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$

(e) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

4. A possible fundamental matrix for the system $X' = \begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix} X$ is $\Phi(t) =$

(a) $\begin{bmatrix} 1 & e^{4t} \\ 0 & -2e^{4t} \end{bmatrix}$

(b) $\begin{bmatrix} 2 & e^{4t} \\ 1 & 2e^{4t} \end{bmatrix}$

(c) $\begin{bmatrix} 1 & e^{4t} \\ 1 & -e^{4t} \end{bmatrix}$

(d) $\begin{bmatrix} 1 & e^{4t} \\ 2 & -2e^{4t} \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 2e^{4t} \\ 2 & -e^{4t} \end{bmatrix}$

5. Suppose that a 2×2 matrix A has the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 1$ and their corresponding eigenvectors $v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, respectively. If $A^2 = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$, then $x + y + z + w =$

(a) -5

(b) -9

(c) -7

(d) -6

(e) -8

6. Let the solution space of the system

$$\begin{aligned} x_1 - 4x_2 - 3x_3 - 7x_4 &= 0 \\ 2x_1 - x_2 + x_3 + 7x_4 &= 0 \\ x_1 + 2x_2 + 3x_3 + 11x_4 &= 0 \end{aligned}$$

have the following vectors $u = (-1, -1, 1, 0)$ and $v = (a, b, 0, 1)$ as basis for the solution space. Then $b - a =$

(a) 3

(b) 2

(c) -2

(d) -1

(e) 1

7. Which of the following sets $s = \{(x_1, x_2, x_3)\}$ forms a subspace of $\mathbb{R}^3$?

(a) All vectors such that $(x_1)^2 + (x_2)^2 = 0$

(b) All vectors such that $x_1 + x_2 + x_3 = 1$

(c) All vectors such that $|x_1| + |x_3| = 1$

(d) All vectors such that $x_1 x_2 = x_3$

(e) All vectors such that $x_2 = 1$

8. Given that $A = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 0 & 1 \\ 2 & -2 & 3 \end{bmatrix}$. If $A^{-1} = aA^2 + bA + cI$, then $12(a + b + c) =$

(a) 11

(b) 9

(c) 12

(d) 8

(e) 10

9. If $X = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} e^{5t}$ is the solution of the initial value problem $X' = \begin{bmatrix} 3 & -2 & 0 \\ -1 & 3 & -2 \\ 0 & -1 & 3 \end{bmatrix} X,\ X(0) = \begin{bmatrix} 0 \\ 2 \\ 6 \end{bmatrix}$, then $c_1^2 + c_2^2 =$

(a) 14

(b) 12

(c) 15

(d) 13

(e) 11

10. The general solution of $X' = \begin{bmatrix} 1 & -3 \\ 3 & 7 \end{bmatrix} X$ is $X =$

(a) $c_1 \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} -3t+1 \\ 2t \end{bmatrix} e^{4t}$

(b) $c_1 \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 3t+1 \\ 3t \end{bmatrix} e^{4t}$

(c) $c_1 \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} -3t+1 \\ 3t \end{bmatrix} e^{4t}$

(d) $c_1 \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 3t+2 \\ 3t \end{bmatrix} e^{4t}$

(e) $c_1 \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} -2t+1 \\ 3t \end{bmatrix} e^{4t}$

11. Let $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix}$. If $e^{At} = \begin{bmatrix} e^{2t} & f(t) & h(t) \\ 0 & e^{2t} & 6te^{2t} \\ 0 & 0 & e^{2t} \end{bmatrix}$, then $f(1)+h(1) =$ (Hint: Consider $A = D+B$, where B is a nilpotent matrix)

(a) $16e^2$

(b) $14e^2$

(c) $18e^2$

(d) $17e^2$

(e) $15e^2$

12. If the general solution of $X' = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} X$ is $X = c_1 \begin{bmatrix} 1 \\ a \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} b \\ 1 \end{bmatrix} e^{\lambda t}$, then $a + b + \lambda =$

(a) 3

(b) 4

(c) 5

(d) 2

(e) 6

13. The general solution of the differential equation

$$y''' + 3y'' - 4y = 0$$

is $y =$

(a) $c_1e^x + c_2e^{-x} + c_3xe^{-x}$

(b) $c_1e^x + c_2e^{-2x} + c_3xe^{-2x}$

(c) $c_1e^{-x} + c_2e^{-2x} + c_3xe^{-2x}$

(d) $c_1e^x + c_2e^{-2x}$

(e) $c_1e^{-x} + c_2e^{2x} + c_3xe^{2x}$

14. The characteristic equation of a matrix A is $(\lambda - 1)(\lambda + 2)^3 = 0$, where we have only two linearly independent eigenvectors corresponding to $\lambda = -2$. The Jordan normal form of A is

(a) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

15. If $y_p = \cos x \ln|\csc x + \cot x| + u_2 \sin x$ is a particular solution of the differential equation $y'' + y = \csc^2 x$, then $u_2(x) =$

(a) $-\sec x$

(b) $-\cot x$

(c) $-\csc x$

(d) $-\tan x$

(e) $-\cos x$

16. Let $F(x, y) = e$ be the solution of the initial-value problem

$$(\cos x + \ln y)\, dx + \left(\frac{x}{y} + e^y\right) dy = 0, \quad y(0) = 1.$$

Then $F(0, 2)$ equals to

(a) e

(b) e^3

(c) e^2

(d) e^{-1}

(e) e^4

17. If y_p is a particular solution of the differential equation $y'' + 4y = 3x^3$, with $y_p(0) = 0$ then $y_p(-1) =$

(a) $\dfrac{-1}{2}$

(b) $\dfrac{1}{2}$

(c) $\dfrac{-3}{8}$

(d) $\dfrac{3}{8}$

(e) $\dfrac{9}{8}$

18. If $y = y(x)$ is the solution of the initial-value problem

$$xy' = 3y + x^4 \cos x, \quad y(2\pi) = 0,$$

then $\left(\dfrac{2}{\pi}\right)^3 y\left(\dfrac{\pi}{2}\right) =$

(a) -2

(b) 1

(c) 3

(d) 2

(e) -1

19. A basis for $\mathbb{R}^2$ which consists of eigenvectors of the matrix $A = \begin{bmatrix} 10 & -8 \\ 6 & -4 \end{bmatrix}$ is

(a) $\left\{ \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

(d) $\left\{ \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\}$

(e) $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$

20. By using Cramer's Rule to solve the system

$$\begin{aligned} 5x_1 + 4x_2 - 2x_3 &= 4 \\ 2x_1 + 3x_3 &= 2 \\ 2x_1 - x_2 + x_3 &= 1 \end{aligned}$$

then $x_3 =$

(a) $\frac{5}{7}$

(b) $\frac{2}{7}$

(c) $\frac{1}{7}$

(d) $\frac{8}{7}$

(e) $\frac{13}{7}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\lambda = 4 \pm 3i$: $X = e^{4t}[(\cos 3t, \sin 3t) - 2(\sin 3t, -\cos 3t)]$, $X(\pi) = (-1,-2)e^{4\pi}$.
2	**(e)**	$x_4' = -x_1 + 3x_2 - 6x_3 + \cos 3t$ gives last row $(-1, 3, -6, 0)$.
3	**(b)**	$e^{-As}F = (2,1)$, so $X_p = e^{At}(2t,t) = (2t,t)$ and $X_p(1) = (2,1)$.
4	**(d)**	$\lambda = 0$: $(1,2)$; $\lambda = 4$: $(1,-2)e^{4t}$; these are the columns of Φ.
5	**(c)**	$A^2 = PD^2P^{-1} = \begin{bmatrix} 13 & -18 \\ 6 & -8 \end{bmatrix}$, sum $= -7$.
6	**(b)**	Plug v into the equations: $a - 4b = 7$, $2a - b = -7$ gives $a = -5$, $b = -3$, so $b - a = 2$.
7	**(a)**	$x_1^2 + x_2^2 = 0$ means $x_1 = x_2 = 0$: the set $\{(0,0,x_3)\}$ is a subspace.
8	**(e)**	$A^3 - 7A^2 + 16A - 12I = 0$, so $12A^{-1} = A^2 - 7A + 16I$ and $12(a+b+c) = 10$.
9	**(d)**	$c_1 + c_2 + c_3 = 0$, $c_1 - c_3 = 1$, $c_1 - c_2 + c_3 = 6$: $c = (2,-3,1)$, so $4 + 9 = 13$.
10	**(c)**	Double $\lambda = 4$, $\mathbf{v} = (-3,3)$, $\mathbf{w} = (1,0)$: second solution $(-3t+1, 3t)e^{4t}$.
11	**(a)**	$e^{At} = e^{2t}(I + Bt + B^2t^2/2)$: $f = 3te^{2t}$, $h = (4t + 9t^2)e^{2t}$, sum at 1 is $16e^2$.
12	**(b)**	$\lambda = -2, 5$; vectors $(1,-3)$ and $(2,1)$: $a = -3$, $b = 2$, $\lambda = 5$, sum 4.
13	**(b)**	$r^3 + 3r^2 - 4 = (r-1)(r+2)^2$: $y = c_1e^x + c_2e^{-2x} + c_3xe^{-2x}$.
14	**(d)**	Two blocks for $\lambda = -2$ (sizes 2 and 1) plus a 1×1 block for $\lambda = 1$.
15	**(c)**	$W = 1$, $u_2' = \cos x \csc^2 x$, so $u_2 = -\csc x$.
16	**(c)**	Exact: $F = \sin x + x \ln y + e^y$, $F(0,1) = e$ and $F(0,2) = e^2$.
17	**(d)**	Undetermined coefficients: $y_p = \frac{3}{4}x^3 - \frac{9}{8}x$, so $y_p(-1) = \frac{3}{8}$.
18	**(b)**	$\rho = x^{-3}$: $x^{-3}y = \sin x$, so $y = x^3 \sin x$ and the value is 1.
19	**(c)**	$\lambda = 2, 4$ with eigenvectors $(1,1)$ and $(4,3)$.
20	**(b)**	$\det A = 35$, $\det A_3 = 10$, so $x_3 = \frac{10}{35} = \frac{2}{7}$.